(†) The rings should be regarded as commutative rings with identity.

Theorem 0.1. *: Let the set Σ be a partially ordered set with the relation $\leqslant$. Then TFAE:*
(1) Every increasing chain in Σ terminates. i.e., for any increasing chain $x_1 \leqslant x_2 \leqslant \cdots \leqslant x_k \leqslant \dots$, $\exists n \in \mathbb{N}$ such that $x_n = x_{n+1} = x_{n+2} = \dots$;
(2) Every nonempty subset of Σ has a maximal element.

Proof: (1)⇒(2): Let $S \subseteq \Sigma$ and $S \neq \phi$. If S has no maximal element, take $\forall a_1 \in S$, then $\exists a_2 \in S$ such that $a_1 \leqslant a_2$ and $a_1 \neq a_2$. Since a_2 is not the maximal element of S, then $\exists a_3 \in S$ such that $a_2 \leqslant a_3$ and $a_2 \neq a_3$. By induction, we get an infinite increasig chain

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n \leqslant \dots$$

but $a_i \neq a_{i+1}$ ($\forall i \in \mathbb{N}$), which contradicts to the condition (1). Thus S has a maximal element;

(2)⇒(1): For any increasing chain

$$a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n \leqslant \dots$$

let $S := \{a_k | \ k = 1, 2, \dots\}$. Since $S \subseteq \Sigma$ and $S \neq \phi$, hence S has a maximal element, i.e., $\exists a_t \in S$ such that $a_i \leqslant a_t$ for $\forall i = 1, 2, \dots$, which implies that $a_t = a_{t+1} = a_{t+2} = \dots$. □

Now, let A be a ring and M is an A-moduole, let $\Sigma := \{N | \ N \subseteq M \text{ is a submodule of } M\}$, then we can know that $(\Sigma, \subseteq)$ and $(\Sigma, \supseteq)$ become partially ordered set.

a.c.c and d.c.c

Definition 0.1. : The A-module M is said to satisfy the **ascending chain condition (a.c.c)** if $(\Sigma, \subseteq)$ satisfies one of the conditions in Theorem 0.1; The A-module M is said to satisfy the **descending chain condition (d.c.c)** if $(\Sigma, \supseteq)$ satisfies one of the conditions in Theorem 0.1.

Noetherian Module and Artinian Module

Definition 0.2. : An A-module M is said to be **Noetherian** if M satisfies a.c.c; An A-module M is said to be **Artinian** if M satisfies d.c.c.

Noetherian Ring and Artinian Ring

Definition 0.3. : If a ring A satisfy a.c.c (resp. d.c.c) as an A-module, then A is called a **Noetherian ring (resp. Artinian ring)**.

Example 0.1. : Any finite abelian group is both Noetherian and Artinian as $\mathbb{Z}$-moduole.

Example 0.2. :For a field k, it is both a Noetherian ring and an Artinian ring.

Example 0.3. : Let A be a PID, then A is Noetherian. But A may not be Artinian.

Example 0.4. : Let p be a fixed prime. Define $G := \{a \in \mathbb{Q}/\mathbb{Z} | \exists n \in \mathbb{N}$ such that $p^n a = 0\}$, a subgroup of $\mathbb{Q}/\mathbb{Z}$. For any $n \in \mathbb{Z}_+$, define $G_n := \{a \in G | p^n a = 0\}$, which is a subgroup of G of order p^n. Then $G_0 \subseteq G_1 \subseteq \cdots \subseteq G_n \subseteq \ldots$ does not terminates, hence G is not Noetherian. Since any subgroup of G is of the form G_n, so G is Artinian.

Example 0.5. : Let k be a field, consider the polynomial ring $k[x_1, x_2, \ldots, x_n]$, then we have two chains $\langle x_1 \rangle \subseteq \langle x_1, x_2 \rangle \subseteq \ldots$ and $\langle x_1 \rangle \supseteq \langle x_1^2 \rangle \supseteq \ldots$, neither of which terminates. Hence $k[x_1, x_2, \ldots, x_n]$ is neither Noetherian nor Artinian.

Theorem 0.2. *: Suppose M is an A-module, then M is Noetherian $\Leftrightarrow$ every submodule of M is finitely generated.*

Proof:($\Rightarrow$): Let N be a submodule of M. Let

$$\Sigma := \{S | \ S \subseteq N, \ S \text{ is a submodule of } M \text{ and } S \text{ is finitely generated}\}$$

hence $(0) \in \Sigma$ and $\Sigma \neq \phi$. Since M is Noetherian, Σ has a maximal element N_0. If $N_0 \neq N$, then $\exists x \in N \setminus N_0$ and hence $N_0 + \langle x \rangle \subseteq N$. Since $N_0 \in \Sigma$ is finitely generated, then $N_0 + \langle x \rangle$ is also finitely generated, i.e., $N_0 + \langle x \rangle \in \Sigma$, which contradicts to the fact that N_0 is the maximal element of Σ. Thus we get $N_0 = N$ and N is finitely generated;

($\Leftarrow$): Let $M_1 \subseteq M_2 \subseteq \cdots \subseteq M_n \subseteq \ldots$ be any increasing chain of submodules of M. Then $N := \bigcup_{k=1}^{\infty} M_k$ is a submoduole of M, hence $N = \bigcup_{k=1}^{\infty} M_k$ is finitely generated. Let $N = \langle x_1, x_2, \ldots, x_r \rangle$, then $x_i \in N$, which means $x_i \in M_{n_i}$ for some n_i $(i = 1, 2, \ldots, r)$. Let $n = max\{n_1, n_2, \ldots, n_r\}$, then $x_i \in M$ $(i = 1, 2, \ldots, r)$. Thus we get $M_n = M_{n+1} = \cdots = N$, which means that M is Noetherian. $\square$

Proposition 0.1. *: Let $0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$ be an exact sequence of A-modules. Then we have*
(1) M is Noetherian $\Leftrightarrow M', M''$ are Noetherian.
(2) M is Artinian $\Leftrightarrow M', M''$ are Artinian.

Proof: (1) ($\Rightarrow$): Since M is Noetherian, let

$$N'_1 \subseteq N'_2 \subseteq \cdots \subseteq N'_k \subseteq \ldots$$

be an increasing chain of submodules of M' and let

$$N''_1 \subseteq N''_2 \subseteq \cdots \subseteq N''_k \subseteq \ldots$$

be an increasing chain of submodules of M''. Hence we know that

$$f(N'_1) \subseteq f(N'_2) \subseteq \cdots \subseteq f(N'_k) \subseteq \ldots$$

is an increasing chain of submodules of M and thus it terminates, i.e., $\exists n \in \mathbb{N}$ such that $f(N'_n) = f(N'_{n+1}) = f(N'_{n+2}) = \dots$. Since f is injecitive, we have $f^{-1}(f(N'_i)) = N'_i$, which implies $N'_n = N'_{n+1} = N'_{n+2} = \dots$, hence M' is Noetherian; similarly, we know that

$$g^{-1}(N''_1) \subseteq g^{-1}(N''_2) \subseteq \dots \subseteq g^{-1}(N''_k) \subseteq \dots$$

is an increasing chain of submodules of M'' and thus it terminates, i.e., $\exists n \in \mathbb{N}$ such that $g^{-1}(N''_n) = g^{-1}(N''_{n+1}) = g^{-1}(N''_{n+2}) = \dots$. Since g is surjective, we have $g(g^{-1}(N''_i)) = N''_i$, which implies that $N''_n = N''_{n+1} = N''_{n+2} = \dots$, hence M'' is Noetherian;

($\Leftarrow$): Let N be a submodule of M. Then $N \cap M'$ is a submoduole of M', hence $N \cap M'$ is finitely generated. Assume that $N \cap M' = \langle x_1, x_2, \dots, x_r \rangle$. Consider $(N + M')/M'$, wchich is a submodule of $M/M' \cong M''$, hence $(N+M')/M'$ is finitely generated. Assume that $\langle \overline{y}_1, \overline{y}_2, \dots, \overline{y}_s \rangle = (N+M')/M' \cong N/(N \cap M')$. Then we assert that

$$N = \langle x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_s \rangle$$

indeed, for $\forall n \in \mathbb{N}$, if $n \in N \cap M'$, then $n \triangleq \sum_{i=1}^r a_i x_i$; if $n \neq N \cap M'$, then $\overline{0} \neq \overline{n} \in (N + M')/M' \cong N/(N \cap M')$, hence $\overline{n} \triangleq \sum_{j=1}^s b_j \overline{y}_j$ and $n - \sum_{j=1}^s b_j y_j \in N \cap M'$, which means that $n - \sum_{j=1}^s b_j y_j \triangleq \sum_{i=1}^r a_i x_i$, hence $n \in \langle x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_s \rangle$ and thus $N \subseteq \langle x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_s \rangle$. It's clear that $\langle x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_s \rangle \subseteq N$. Thus $N = \langle x_1, x_2, \dots, x_r, y_1, y_2, \dots, y_s \rangle$, i.e., N is finitely generated and thus M is Noetherian;

(2) ($\Rightarrow$):Similar to the proof of (1);

($\Leftarrow$): Suppose M', M'' are Artinian. Let

$$N_1 \supseteq N_2 \supseteq \dots \supseteq N_k \supseteq \dots$$

be a decreasing chain of submodules of M, hence

$$M' \cap N_1 \supseteq M' \cap N_2 \supseteq \dots \supseteq M' \cap N_k \supseteq \dots$$

is a decreasing chain of submodules of M', hence it terminates, i.e., $\exists n_1 \in \mathbb{N}$ such that $M' \cap N_{n_1} = M' \cap N_{n_1+1} = M' \cap N_{n_1+2} = \dots$. Since $M'' \cong M/M'$, thus

$$(N_1 + M')/M' \supseteq (N_2 + M')/M' \supseteq \dots \supseteq (N_k + M')/M' \supseteq \dots$$

is a decreasing chain of submodules of M'', thus it terminates. Since $(N_i + M')/M' \cong N_i/(M' \cap N_i)$, $\exists n_2 \in \mathbb{N}$ such that

$$N_{n_2}/(M' \cap N_{n_2}) = N_{n_2+1}/(M' \cap N_{n_2+1}) = N_{n_2+2}/(M' \cap N_{n_2+2}) = \dots$$

let $n_0 = max\{n_1, n_2\}$, then we get

$$M' \cap N_{n_0} = M' \cap N_{n_0+1} = \dots$$

$$N_{n_0} = (M' \cap N_{n_0}) = N_{n_0+1}/(M' \cap N_{n_0+1}) = \dots$$

hence we have

$$N_{n_0} = N_{n_0+1} = N_{n_0+2} = \dots$$

which means that M is Artinian. $\square$

Corollary 0.1. *: If $M_1, M_2, \ldots, M_n$ are Noetherian (resp. Artinian), then so is $\bigoplus_{i=1}^n M_i$.*

Proof: Consider the exact sequence

$$0 \longrightarrow M_1 \longrightarrow M_1 \bigoplus M_2 \longrightarrow M_2 \longrightarrow 0$$

so $M_1 \bigoplus M_2$ is Noetherian (resp. Artinian), then consider the exact sequence

$$0 \longrightarrow M_3 \longrightarrow \bigoplus_{i=1}^3 M_i \longrightarrow M_1 \bigoplus M_2 \longrightarrow 0$$

so $\bigoplus_{i=1}^3 M_i$ is Noetherian (resp. Artinian). By induction, we know that $\bigoplus_{i=1}^n M_i$ is Noetherian (resp. Artinian). $\square$

Corollary 0.2. *: If A is Noetherian (resp. Artinian) and M is a finitely generated A-module, then M is Noetherian (resp. Artinian).*

Proof: Recall that M is a finitely generated A-module $\Leftrightarrow \exists n \in \mathbb{N}$ and $\exists$ an epimorphism $A^{(n)} \to M$, where $A^{(n)} = A \oplus A \oplus \cdots \oplus A$. Thus A is Noetherian (resp. Artinian) $\Rightarrow A(n)$ is Noetherian (resp. Artinian), hence M is Noetherian (resp. Artinian). $\square$

Corollary 0.3. *: If A is Noetherian (resp. Artinian) and $I \subseteq A$ is an ideal of A, then A/I is Noetherian (resp. Artinian).*

Proof: Consider the exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

By Proposition 0.1, the conclusion holds. $\square$

Remark 0.1. *: For a Noetherian (resp. Artinian) A-module M, by the exact sequnce*

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

we know that both the submodule N and the quotient module M/N are Noetherian (resp. Artinian).